

# Titanic - Machine Learning

## (part 2)

```
In [11]: # =====
# 1. DATA LOAD
# =====
import pandas as pd

url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df = pd.read_csv(url)

print(df)

df.columns = df.columns.str.strip()

df.columns = df.columns.str.lower()

print(df.columns)
# =====
# 2. CLEANING
# =====
df = df[['survived', 'pclass', 'age', 'sex', 'embarked']].head()

df['age'] = df['age'].fillna(df['age'].median())
df['sex'] = df['sex'].map({'male':0, 'female':1})
df['embarked'] = df['embarked'].fillna('S')
df['embarked'] = df['embarked'].map({'S':0, 'C':1, 'Q':2})

# =====
# 3. TRAIN TEST SPLIT
# =====
from sklearn.model_selection import train_test_split

trup=['pclass', 'age', 'sex', 'embarked']
X = df[trup]
y = df['survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# =====
# 4. LOGISTIC REGRESSION
# =====
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

log_model = LogisticRegression()
log_model.fit(X_train, y_train)
log_pred = log_model.predict(X_test)

print("Logistic Accuracy:", accuracy_score(y_test, log_pred))

# =====
# 5. DECISION TREE
# =====
from sklearn.tree import DecisionTreeClassifier

tree_model = DecisionTreeClassifier()
tree_model.fit(X_train, y_train)
tree_pred = tree_model.predict(X_test)

print("Tree Accuracy:", accuracy_score(y_test, tree_pred))

# =====
# 6. SCALING
# =====
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2)

# =====
```

```
# 7. LOGISTIC AFTER SCALING
# =====
log_model2 = LogisticRegression()
log_model2.fit(X_train, y_train)
log_pred2 = log_model2.predict(X_test)

print("Scaled Logistic Accuracy:", accuracy_score(y_test, log_pred2))

# =====
# 8. METRICS
# =====
from sklearn.metrics import precision_score, recall_score, f1_score

print("Precision:", precision_score(y_test, log_pred))
print("Recall:", recall_score(y_test, log_pred))
print("F1 Score:", f1_score(y_test, log_pred))

# LogisticRegression (Underfitting tendency)
log_model = LogisticRegression()
log_model.fit(X_train, y_train)
print("Logistic Train Accuracy:", accuracy_score(y_train, log_model.predict(X_train)))
print("Logistic Test Accuracy:", accuracy_score(y_test, log_model.predict(X_test)))

# Decision Tree (Overfitting tendency)
tree_model = DecisionTreeClassifier()
tree_model.fit(X_train, y_train)
print("Tree Train Accuracy:", accuracy_score(y_train, tree_model.predict(X_train)))
print("Tree Test Accuracy:", accuracy_score(y_test, tree_model.predict(X_test)))
```

|     | PassengerId | Survived | Pclass | \ |
|-----|-------------|----------|--------|---|
| 0   | 1           | 0        | 3      |   |
| 1   | 2           | 1        | 1      |   |
| 2   | 3           | 1        | 3      |   |
| 3   | 4           | 1        | 1      |   |
| 4   | 5           | 0        | 3      |   |
| ..  | ...         | ...      | ...    |   |
| 886 | 887         | 0        | 2      |   |
| 887 | 888         | 1        | 1      |   |
| 888 | 889         | 0        | 3      |   |
| 889 | 890         | 1        | 1      |   |
| 890 | 891         | 0        | 3      |   |

|     | Name                                              | Sex    | Age  | SibSp | \ |
|-----|---------------------------------------------------|--------|------|-------|---|
| 0   | Braund, Mr. Owen Harris                           | male   | 22.0 | 1     |   |
| 1   | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0 | 1     |   |
| 2   | Heikkinen, Miss. Laina                            | female | 26.0 | 0     |   |
| 3   | Futrelle, Mrs. Jacques Heath (Lily May Peel)      | female | 35.0 | 1     |   |
| 4   | Allen, Mr. William Henry                          | male   | 35.0 | 0     |   |
| ..  | ...                                               | ...    | ...  | ...   |   |
| 886 | Montvila, Rev. Juozas                             | male   | 27.0 | 0     |   |
| 887 | Graham, Miss. Margaret Edith                      | female | 19.0 | 0     |   |
| 888 | Johnston, Miss. Catherine Helen "Carrie"          | female | NaN  | 1     |   |
| 889 | Behr, Mr. Karl Howell                             | male   | 26.0 | 0     |   |
| 890 | Dooley, Mr. Patrick                               | male   | 32.0 | 0     |   |

|     | Parch | Ticket           | Fare    | Cabin | Embarked |
|-----|-------|------------------|---------|-------|----------|
| 0   | 0     | A/5 21171        | 7.2500  | NaN   | S        |
| 1   | 0     | PC 17599         | 71.2833 | C85   | C        |
| 2   | 0     | STON/O2. 3101282 | 7.9250  | NaN   | S        |
| 3   | 0     | 113803           | 53.1000 | C123  | S        |
| 4   | 0     | 373450           | 8.0500  | NaN   | S        |
| ..  | ...   | ...              | ...     | ...   | ...      |
| 886 | 0     | 211536           | 13.0000 | NaN   | S        |
| 887 | 0     | 112053           | 30.0000 | B42   | S        |
| 888 | 2     | W./C. 6607       | 23.4500 | NaN   | S        |
| 889 | 0     | 111369           | 30.0000 | C148  | C        |
| 890 | 0     | 370376           | 7.7500  | NaN   | Q        |

```
[891 rows x 12 columns]
Index(['passengerid', 'survived', 'pclass', 'name', 'sex', 'age', 'sibsp',
      'parch', 'ticket', 'fare', 'cabin', 'embarked'],
      dtype='object')
Logistic Accuracy: 1.0
Tree Accuracy: 1.0
Scaled Logistic Accuracy: 0.0
Precision: 0.0
Recall: 0.0
F1 Score: 0.0
Logistic Train Accuracy: 1.0
Logistic Test Accuracy: 0.0
Tree Train Accuracy: 1.0
Tree Test Accuracy: 1.0
```

```
C:\Users\RAMIZ\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\metrics\_classification.py:1833
: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 due to no true samples. Use `zero_division`
parameter to control this behavior.
_warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
```

In [ ]: